

Model distg, Experiment *real_data* – *run1* – 1031 – 1943

November 3, 2025

1 Model Description and Training

The distg is used with a channel dimension of 64. The model has an overall number of 2690192 of learnable parameters. The model was trained for 80 epochs with a batch size of 64. The adam optimizer was used with an initial learning rate of 0.0002. The learning rate is decreased every 15 epochs by a factor of 0.5.

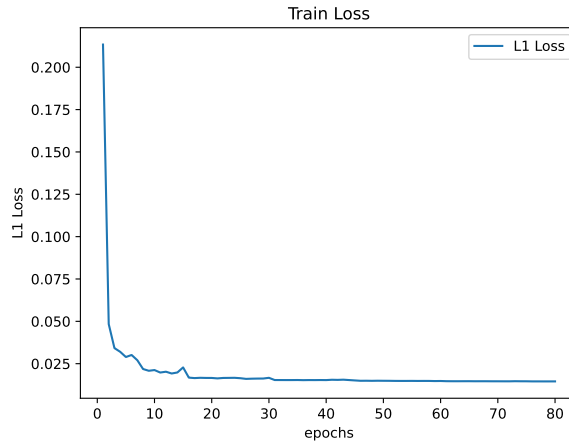


Figure 1: Train Loss

2 Qualitative Results

3 Quantitative Results

3.1 PSNR

	Lab day	Lab night	Indoors day	Indoors night	Showroom	Outdoors
MAX	34.224	36.134	34.972	39.482	32.644	29.731
LAST	34.210	36.112	34.963	39.428	32.634	29.731

3.2 SSIM

	Lab day	Lab night	Indoors day	Indoors night	Showroom	Outdoors
MAX	0.866	0.877	0.843	0.920	0.869	0.825
LAST	0.866	0.876	0.843	0.919	0.868	0.824

3.3 SAM

	Lab day	Lab night	Indoors day	Indoors night	Showroom	Outdoors
MIN	0.075	0.088	0.053	0.189	0.080	0.094
LAST	0.075	0.088	0.053	0.190	0.080	0.094

3.4 SRE

	Lab day	Lab night	Indoors day	Indoors night	Showroom	Outdoors
MIN	8.552	7.381	8.013	4.410	10.354	13.350
LAST	8.557	7.389	8.013	4.429	10.357	13.350



Figure 2: Qualitative Results, from left to right, Bicubically upsampled, Super Resolved, Original Data

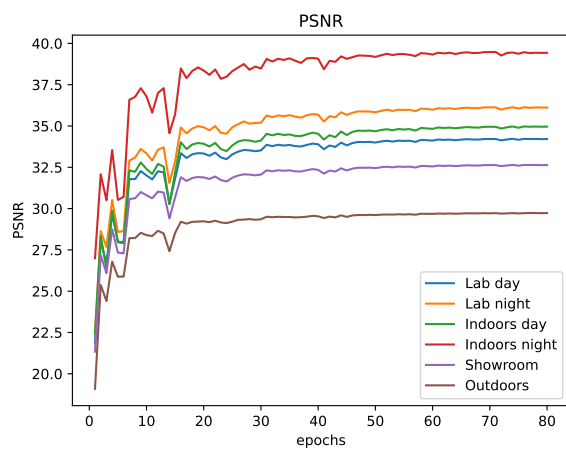


Figure 3: PSNR

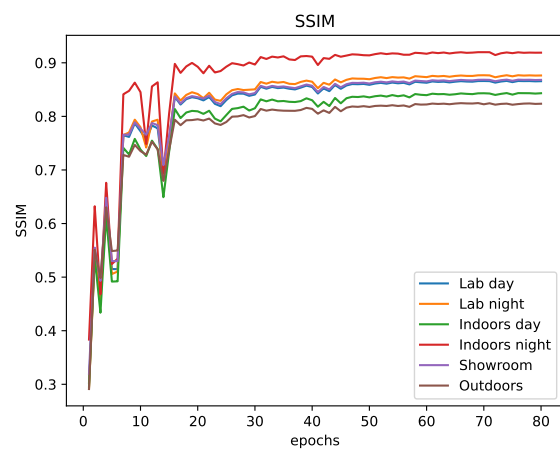


Figure 4: SSIM

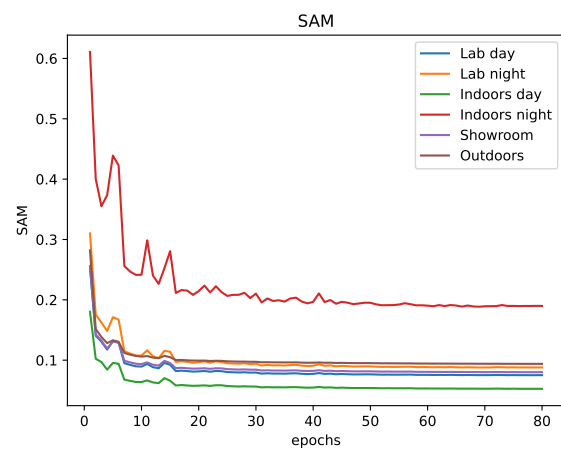


Figure 5: SAM

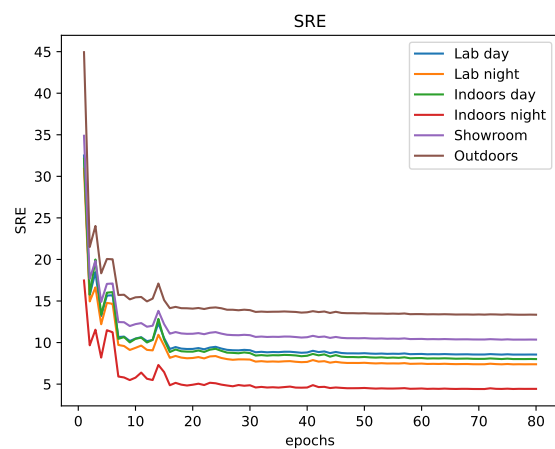


Figure 6: SRE